Material cases: evidence and mechanisms

Four case families connect switching physics to measurements: SiO_2 conductive-bridge RAM, TaO_x RESET statistics, HfO_2/Ti aging, and two- and three-terminal atomic switches. Values below are either explicitly identified as historical observations or stated as hypothetical teaching models.

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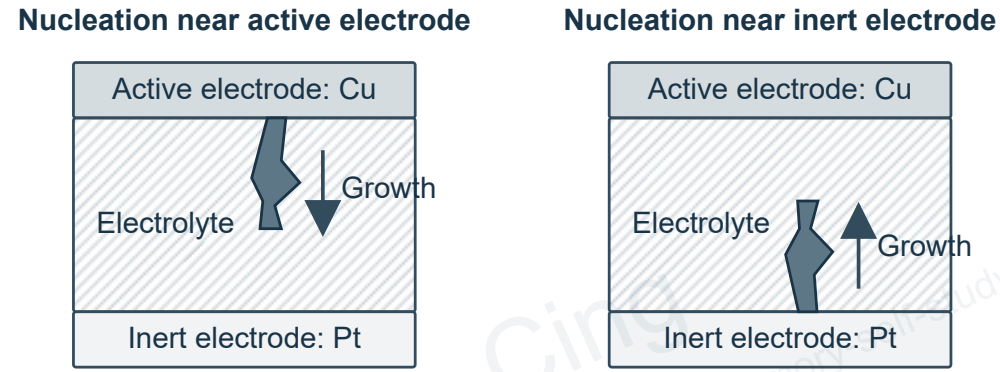
Compare mechanisms through their evidence

Case	Primary observation	A proposed explanation	What needs checking
SiO_2 CBRAM	switching and transport depend on bias and environment	metal ions participate in a conductive bridge	composition, polarity, transport and interfaces
TaO_x RESET	distributions differ with measurement and initial state	thermally assisted local rupture	current, temperature model and sample variability
HfO_2/Ti	the HRS/LRS window evolves during storage	oxygen redistribution and Ti oxidation	separate devices, aging protocol and depth-profile artifacts
Atomic switch	a conducting gap can be opened or closed	nucleation, growth and dissolution	field, pulse timing and alternative kinetics

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A conductive bridge is a process, not a single reaction

Conductive bridge: nucleation can differ
Schematic; reaction, supply and transport govern growth



Both scenarios are possible; polarity alone does not set direction.
Bridge formation also needs ion supply, reduction and connection.

For an illustrative Cu-active electrode, oxidation can supply ions: $\mathrm{Cu} \rightarrow \mathrm{Cu}^{+} + e^{-}$. A reduction step can return ions to metal: $\mathrm{Cu}^{+} + e^{-} \rightarrow \mathrm{Cu}$. Their locations and rates depend on electrode materials, interfacial kinetics, ion transport and electronic access.

Polarity alone does not determine morphology

Ion drift, diffusion, interfacial reduction and existing clusters can compete. Published material systems show growth toward either electrode or distributed nucleation; a voltage sign by itself cannot establish a universal growth direction. The return of HRS also need not remove every metallic atom.

To test a mechanism, align electrical switching with composition-sensitive imaging, repeat cycles, control humidity and temperature, and check whether the probe changes the state.

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A diffusion scale is not a SET-time prediction

For ideal one-dimensional unbiased diffusion from a point, the mean-square displacement is $\langle x^2 \rangle = 2Dt$. For a deliberately hypothetical $D = 10^{-21}\,\mathrm{cm^2/s}$ and rms displacement $20\,\mathrm{nm} = 2 \times 10^{-6}\,\mathrm{cm}$,

$$t = \frac{(2 \times 10^{-6})^2}{2 \times 10^{-21}} = 2 \times 10^{9}\,\mathrm{s} \approx 63\,\mathrm{yr}.$$

This extrapolation excludes electric-field drift, interfacial reactions and thin-film pathways. It is not a device switching time or evidence identifying one mechanism.

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Field and humidity change the boundary conditions

Under a field, ionic flux has drift and diffusion contributions; a high-field empirical fit applies only within its measured range. Boundary reaction rates and available ions also matter.

In a particular reported Cu/SiO_2/Pt experiment, changing water-vapor conditions affected the redox and ion-population interpretation; this does not establish that humidity always increases ionic mobility. Compare matched samples and histories before attributing a change to one parameter.

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Field drift: a limited numerical comparison

For a hypothetical dilute, isothermal, singly charged carrier in the linear-response regime, use the Einstein relation $\mu = D/(k_B T/e)$ and drift speed $v_d = \mu E$. Take $D = 10^{-21}\,\mathrm{cm^2/s}$, $k_B T/e = 0.02585\,\mathrm{V}$, and $E = 10^5\,\mathrm{V/cm}$:

$$\mu = 3.868 \times 10^{-20}\,\mathrm{cm^2/(V\,s)}, \quad v_d = 3.868 \times 10^{-15}\,\mathrm{cm/s}.$$

The drift distance divided by v_d for $L = 20\,\mathrm{nm}$ is $5.17 \times 10^8\,\mathrm{s}$. This is an illustrative drift scale, not a first-passage prediction or a SET time. Linear response and the Einstein relation are explicit assumptions; the field value by itself does not establish them. For a hopping picture, $qaE \ll k_B T$ is one condition to check. Constant field, dilute independent carriers and these relations may fail in a real switching gap.

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What changing humidity can and cannot identify

In a uniform drift-only illustrative model $J = qn\mu E$. The same change in J could result from a change in mobile-carrier concentration n, mobility μ, or field E; interface reactions and diffusive flux add further ambiguity. A current change alone cannot separately identify n and μ.

At fixed geometry and temperature, measure a concentration-sensitive signal and independently test transport across several fields and matched humidity histories; the result remains conditional on the chosen transport and contact model.

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A simple impedance model and its limits

Suppose a resistance R and capacitance C are in parallel. Their admittance is $Y = 1/R + j\omega C$; hence

$$Z = \frac{R}{1 + j\omega RC}, \qquad -\operatorname{Im} Z = \frac{\omega R^2 C}{1 + (\omega RC)^2}.$$

The imaginary component reaches its maximum at $\omega RC = 1$. Fitting this equivalent circuit describes a response under its assumptions; it does not uniquely reveal a filament's shape or composition.

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Work through the RC scale

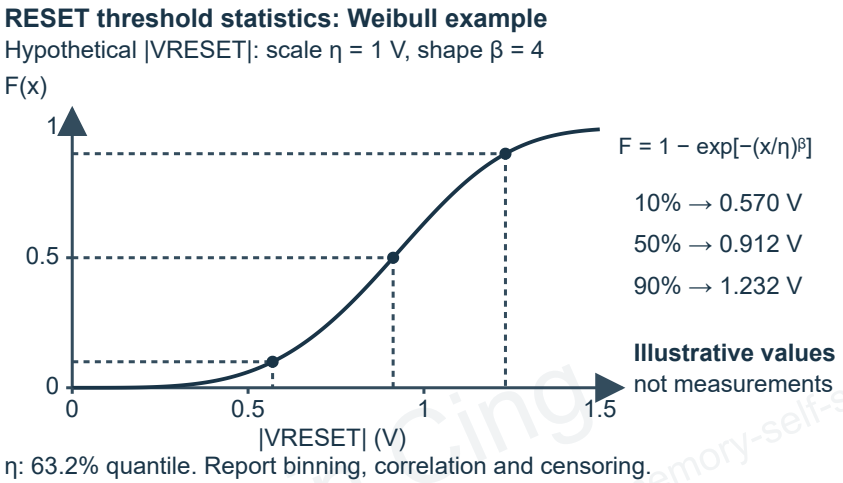
Let $R = 1\,\mathrm{M\Omega}$ and $C = 10\,\mathrm{pF}$ in the parallel model. Then $\tau = RC = 10\,\mu\mathrm{s}$ and

$$f_{\mathrm{peak}} = \frac{1}{2\pi RC} \approx 15.9\,\mathrm{kHz}.$$

Measure over a sufficient frequency range and check lead and electrode contributions. More than one physical arrangement may fit the same spectrum.

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RESET is a distribution, not one universal number



One historical Ta/TaO_x/Pt study used an approximately 18 nm oxide and repeated quasi-DC cycles on a device. SET compliance, RESET stop voltage, voltage polarity, read voltage and cycle history are part of that particular dataset, not universal settings.

For a negative-polarity RESET experiment, define the nonnegative variable $X = |V_{\mathrm{RESET}}|$ before fitting. A peak-current criterion or another chosen criterion must be stated explicitly.

One TaO_x RESET study: protocol before inference

In the specific historical Ta/TaO_x/Pt experiment, the oxide was about 18 nm; approximately 1,000 quasi-DC cycles were recorded on the same device. SET used positive bias with $4\,\mathrm{mA}$ compliance, RESET swept negative to a $-1.5\,\mathrm{V}$ stop, and R_{ON} was read at $0.1\,\mathrm{V}$; the voltage-step time was $1\,\mathrm{ms}$. RESET voltage was defined through the current peak.

These settings and the chosen event definition describe that dataset. Cycle history and resistance binning affect the observed distribution and are not independent population statistics.

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Six resistance bins under a simple assumption

Assume six hypothetical resistance bins and a common peak magnitude $|V_{\mathrm{RESET}}| = 0.9\,\mathrm{V}$. If the instantaneous peak is approximated by $I_{\mathrm{peak}} = |V|/R_{\mathrm{ON}}$:

R_{ON} (Ω)	100	150	200	250	300	400
I_{peak} (mA)	9	6	4.5	3.6	3	2.25

For example, $0.9/150 = 0.006\,\mathrm{A} = 6\,\mathrm{mA}$. These hypothetical fixed-voltage, Ohmic values are not a reconstruction of the published Weibull distribution. In this model the current is inversely proportional to resistance; evolving resistance or a nonconstant peak voltage invalidates the simple calculation.

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Weibull model: CDF and quantile

For $x \geq 0$, shape $\beta > 0$, and scale $\eta > 0$, a two-parameter Weibull model is

$$F(x) = 1 - e^{-(x/\eta)^{\beta}}, \qquad x_p = \eta[-\ln(1-p)]^{1/\beta}.$$

At $x = \eta$, $F(\eta) = 1 - e^{-1} \approx 0.632$. This is a model definition; fitting data does not establish that all devices obey it. Use magnitude consistently and treat censored observations explicitly.

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Weibull worked example

Assume $\eta = 1\,\mathrm{V}$, $\beta = 4$. At $0.8\,\mathrm{V}$, $(0.8)^4 = 0.4096$ and $F = 1 - e^{-0.4096} \approx 0.3361$.

The 10%, 50%, and 90% quantiles are approximately 0.5697, 0.9124, and $1.2318\,\mathrm{V}$. They describe the assumed distribution, not a measured performance guarantee.

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How to fit without manufacturing certainty

For $x > 0$ and $0 < F < 1$, $\ln[-\ln(1-F)] = \beta \ln x - \beta \ln \eta$. A Weibull plot can reveal departures from a straight line; it cannot validate a mechanism on its own.

Report the sample count, device count, cycle dependence, selection or resistance binning, censoring and confidence intervals. Repeated cycles from one device are not automatically independent devices.

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A thermal RESET model is conditional

Assume steady local heating with constant R and thermal resistance R_{th} to a bath at T_0:

$$P = \frac{V^2}{R}, \qquad T = T_0 + R_{\mathrm{th}} P.$$

This lumped model excludes thermal transients, spatial temperature gradients, evolving resistance and nonthermal mechanisms. Agreement with it can support a hypothesis but cannot uniquely establish it.

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Thermal estimate and scaling

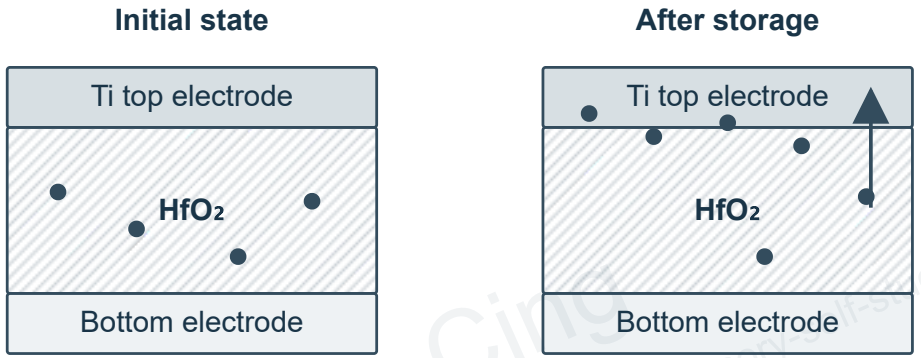
Take hypothetical $R = 150\,\Omega$, $V = 0.9\,\mathrm{V}$, $R_{\mathrm{th}} = 3 \times 10^{4}\,\mathrm{K/W}$. Then $P = 0.81/150 = 5.4\,\mathrm{mW}$ and $\Delta T = 162\,\mathrm{K}$.

If RESET requires the same critical temperature rise, $V_{\mathrm{RESET}}^{2} = \Delta T_c R / R_{\mathrm{th}}$. A nearly constant RESET voltage across different R follows only if ΔT_c and R/R_{th} remain approximately fixed.

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Aging: a measurement protocol is part of the result

HfO_2/Ti ageing: material distribution and evidence
Conceptual illustration; not a measured concentration profile



- Oxygen-related species; arrow shows one possible change.

XPS is indirect; sputtering and sample variation affect interpretation.
Report storage time, structure and measurement conditions.

The cited HfO_2/Ti work studied Pt/HfO_2/Ti/Pt stacks and compared nominal Ti thicknesses, aging intervals and electrical states. The published curves concern specified batches and read conditions; they are not a lifetime guarantee for all HfO_2 cells.

Interpret a changing memory window

For a specified group, a reported HRS/LRS ratio fell from roughly 10^4 to 10^2 over about two weeks: the ratio is 100 times smaller. That does not determine the absolute HRS or LRS values individually.

Depth-profile XPS can support changes in elemental or chemical distributions, but sputtering and sample preparation may alter the profile. It is not a real-time movie of oxygen motion in the same operating cell.

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An aging comparison must preserve the sample identity

The reported stack was Pt/HfO_2 (about 10 nm)/Ti/Pt. Compare nominal Ti thicknesses 10, 30 and 100 nm; the approximately two-week fall from 10^4 to 10^2 concerns the 30 nm group under its particular protocol. The study first applied 15 cycles and then averaged five cycles per cell; storage comparisons used separate fresh devices from the same batch.

Step	What is compared	What the comparison does not establish
Prepare	Same-batch devices; specified Ti thickness and storage time	One device followed continuously unless explicitly measured
Condition and read	Forming and cycles under specified voltage/compliance and read convention	Equal initial states or identical HRS and LRS separately
Profile	Day 1 versus day 60 XPS depth profiles	Undisturbed, real-time motion during switching

At about 60 days, reported ratios for nominal 10/30/100 nm groups were approximately $5 \times 10^4/10^2/10$, respectively, with device and protocol dependence. Read conventions and some labels differ across adjacent source descriptions; do not merge ambiguous conditions without checking the original experiment.

Keep observations and explanations separate

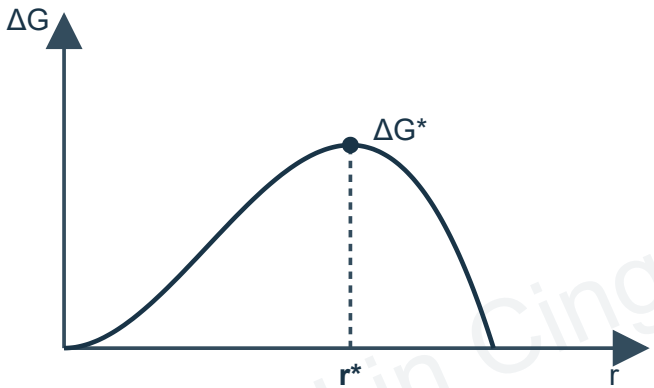
Compare matched fresh and aged devices, specify the Ti thickness, environment, preconditioning, read polarity and uncertainty. A TiO_2 insertion layer is a distinct device structure, not a continuation of a bare-Ti sample.

Reported figure labels and adjacent descriptions may differ in thickness or read conventions. Do not merge uncertain conditions: check the original experiment before reproducing or comparing its numeric values.

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A nucleation model for an atomic switch

Heterogeneous nucleation: surface versus volume energy
Model: fixed interfacial energy, driving force and cap shape



Small: surface cost

Large: volume gain

$r^* = 2\gamma / |\Delta g_v|$

$\Delta G^* = f(\theta)\, 16\pi\gamma^3 / [3|\Delta g_v|^2]$

Barrier ≠ rate: prefactors and transport also matter.
Nanoscale clusters or dynamic fields can invalidate the model.

In a classical spherical-cap approximation, let $\gamma > 0$ be surface energy, $|\Delta g_v| > 0$ the magnitude of bulk driving free energy per volume, and $0 < f(\theta) \le 1$ a geometric factor. Then

$$\Delta G(r) = f(\theta)\left(4\pi r^2\gamma - \frac{4\pi}{3} r^3 |\Delta g_v|\right).$$

Derive the critical radius and barrier

For fixed $f(\theta) > 0$, differentiate with respect to r: $d\Delta G/dr = f(8\pi r\gamma - 4\pi r^2|\Delta g_v|)$. The nonzero stationary radius is

$$r^* = \frac{2\gamma}{|\Delta g_v|}, \qquad \Delta G^* = f(\theta)\frac{16\pi\gamma^3}{3|\Delta g_v|^2}.$$

It is a maximum under this model. Classical continuous geometry and constant material properties may be poor approximations for a cluster only a few atoms wide.

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Nucleation example: calculate before interpreting

Assume $\gamma = 0.03\,\mathrm{J/m^2}$, $|\Delta g_v| = 3 \times 10^7\,\mathrm{J/m^3}$ and $f = 0.1$. Then $r^* = 2 \times 10^{-9}\,\mathrm{m} = 2\,\mathrm{nm}$ and $\Delta G^* \approx 5.03 \times 10^{-20}\,\mathrm{J} \approx 0.314\,\mathrm{eV}$.

At 300 K, use $k_B T \approx 0.02585\,\mathrm{eV}$ and $e^{-\Delta G^*/k_B T} \approx 5.3 \times 10^{-6}$. This factor alone is not a nucleation rate: the attempt frequency, available sites and nonequilibrium transport are not given.

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Time-resolved switching needs a defined event

A voltage edge applied to a device and its parasitic capacitance can produce $I_C = C\,dV/dt$ before a conducting bridge forms. A later jump in conduction current is a different event.

Define SET time relative to the applied pulse and a stated onset criterion; bandwidth, trigger delay and capacitance affect the measurement. A published nanosecond value belongs to its particular structure and measurement chain.

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Separate a capacitive peak from a conducting event

Assume $C = 0.2\,\mathrm{pF}$ and $dV/dt = 1\,\mathrm{V/ns}$. Before conduction, the ideal capacitive current is $0.2\,\mathrm{mA}$; seeing this at $0.5\,\mathrm{ns}$ does not establish SET.

If the stated onset of conduction occurs at $1.2\,\mathrm{ns}$ after pulse start, the defined SET time is $1.2\,\mathrm{ns}$. Device energy requires $\int V(t) I_{\mathrm{device}}(t)\,dt$ with a defined interval and treatment of parasitic energy; a peak alone is insufficient.

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A slope that does not identify a mechanism

Consider two deliberately simplified models on the voltage window $1\text{–}2\,\mathrm{V}$. A nucleation ansatz has $t = A\exp[\Delta G(V)/(k_B T)]$, with $\Delta G(V) = \Delta G_0 - 0.05\,\mathrm{eV}\,(V/1\,\mathrm{V})$, $k_B T = 0.025\,\mathrm{eV}$, and $\Delta G_0 > 0.10\,\mathrm{eV}$ so the barrier remains positive throughout this interval. Choose the combined prefactor so that $\ln(t/\mathrm{ns}) = 6 - 2(V/1\,\mathrm{V})$.

Alternatively, an assumed high-field migration law $v = v_0\exp[eaV/(2k_B T L)]$, with $a = 0.3\,\mathrm{nm}$, $L = 3\,\mathrm{nm}$, and $L/v_0 = e^6\,\mathrm{ns}$, gives the same expression for $t = L/v$. Each law has its own assumptions; neither is established by the example.

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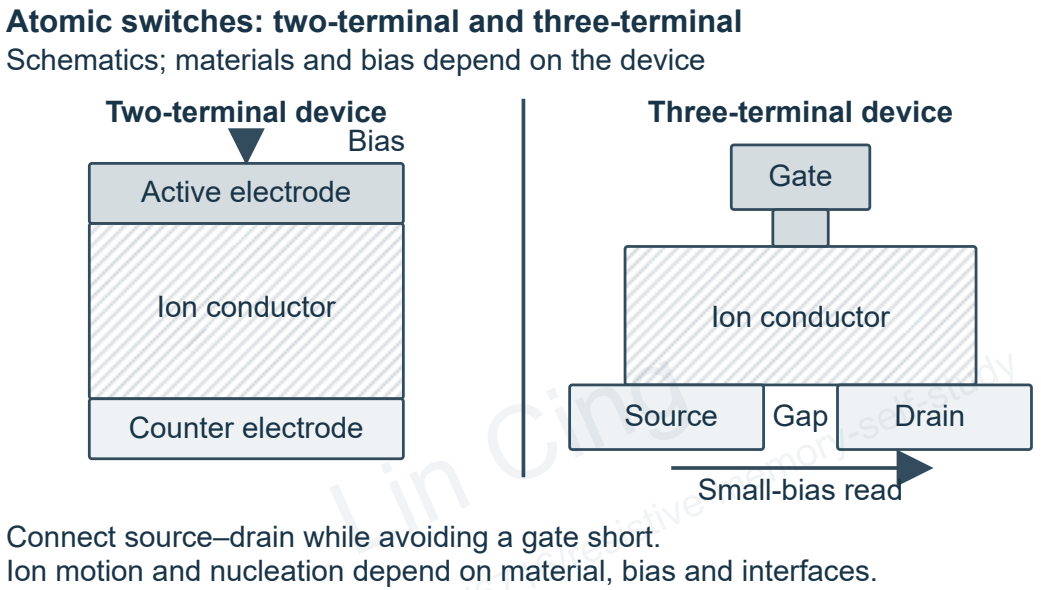
Evaluate competing timing models

For either assumed law, $V = 1$, 1.5, and $2\,\mathrm{V}$ yield $t = e^{4}$, e^{3}, and $e^{2}\,\mathrm{ns}$: approximately 54.6, 20.1, and $7.39\,\mathrm{ns}$.

The identical fitted slope in this constructed example cannot distinguish mechanisms. Compare temperature, gap thickness, composition and time-resolved electrical/structural observations while considering model validity, parasitics and parameter uncertainty. Do not extrapolate a limited high-field fit to zero bias.

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Two terminals and three terminals answer different questions



A two-terminal switch shares the applied voltage between the active electrode, electrolyte and counterelectrode. In a three-terminal example, a gate controls metal-ion supply while a smaller source–drain bias probes the gap. The conducting source–drain path must be distinguished from an unwanted short to gate.

Do not confuse a fit with a unique mechanism

Different kinetic models can fit a limited voltage dependence of switching time. Check their parameter values, geometry, temperature, pulse bandwidth and sensitivity to alternative assumptions. A fitted effective hopping distance larger than an atomic spacing may challenge one model, but a slope alone does not prove the remaining model.

As a final check, separate observed waveform, fitted model, inferred material change and untested predictions.

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Programming current and multilevel states

For a reported Cu/SiO_2 CBRAM family, ON resistance decreases as SET compliance current rises; a fitted form is $R_{\mathrm{ON}} = A / I_{\mathrm{prog}}^{n}$, with exponent near one **for that sample and fit range**. Compliance limits the SET current; it does not guarantee an identical filament geometry on every cycle.

Multiple levels are useful only when the read-current distributions remain distinguishable after programming variation, noise and retention drift. Four labels encode at most two bits per cell; overlapping distributions or time-dependent drift can reduce usable information.

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Trace three hypothetical current levels

Assume the special case $n = 1$ and $A = 0.5\,\mathrm{V}$; these are teaching values, not reported measurements. For $I_{\mathrm{prog}} = 5, 10, 20\,\mu\mathrm{A}$, the model yields $R_{\mathrm{ON}} = 100, 50, 25\,\mathrm{k\Omega}$.

At a non-disturbing $0.1\,\mathrm{V}$ Ohmic read, the corresponding currents are $1, 2, 4\,\mu\mathrm{A}$. These are three nominal levels; to choose thresholds, measure each level's full distribution under the intended read and waiting conditions.

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A conductance scale is not a geometry test

For spin-degenerate ideal channels, $G_0 = 2e^2/h \approx 77.5\,\mu\mathrm{S}$, and $1/G_0 \approx 12.9\,\mathrm{k}\Omega$. A resistance near this scale is suggestive of atomic-scale transport, but $G = G_0 \sum_i T_i$ also allows different channel transmissions to give the same conductance.

The book discusses filament widening, multiple paths and a residual gap as interpretations in different regimes. A resistance threshold by itself cannot prove whether a filament is complete, measure its width, or identify a unique atomic configuration.

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Multilevel retention is a distribution over time

For one reported PECVD Cu/SiO_2 device study, both HRS and LRS were reported to remain near their programmed values for over $10^4\ \mathrm{s}$ at $150\ ^\circ\mathrm{C}$ under that experiment's conditions. This does not establish ten-year retention at another temperature or a bound on an unmeasured tail.

A concentration gradient and interfacial reactions can destabilize a metallic path; high temperature or residue can also change HRS in a particular system. Compare read distributions at stated times, temperatures, read bias and sampling intervals, including failed or censored cells.

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Irradiation: match cell state, dose and sampling

The cited Cu/SiO_2 CBRAM study used stepwise ^{60}Co gamma exposure up to $7.1\,\mathrm{Mrad(SiO_2)}$. Virgin measurements included three $10 \times 10\,\mu\mathrm{m}^2$ and three $100 \times 100\,\mu\mathrm{m}^2$ cells, with sweeps limited to $100\,\mathrm{mV}$ to reduce electrode oxidation during this measurement.

For switched-state distributions, ten $10 \times 10\,\mu\mathrm{m}^2$ cells were cycled ten times at different doses. Similar plotted HRS/LRS distributions support limited sensitivity within that tested set, not universal radiation immunity or an upper bound on rare failures.

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Irradiation and multilevel programming

The same chapter reports three $10 \times 10\,\mu\mathrm{m}^2$ cells programmed over a range of compliance currents before and after the $7.1\,\mathrm{Mrad(SiO_2)}$ exposure; the plotted resistance–current curves were similar under those conditions.

The dose unit includes the reference material: do not silently replace $\mathrm{Mrad(SiO_2)}$ with a different material's absorbed-dose value. Compare the whole state distribution, dose history, read bias and repeated-cell dependence; three cells do not quantify low-probability failures.

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Humidity-controlled cyclic voltammetry: a separate experiment

One reported Cu/Ta_2O_5/Pt cyclic-voltammetry study varied RH from 5% to 85% using dry and humidified nitrogen and used a $100\,\mathrm{mV/s}$ sweep for the plotted RH comparison. With increasing RH, oxidation and reduction peaks became clearer; the reported reduction-peak position changed up to around 22% RH and showed little further change at higher RH in that comparison.

The inferred ion concentration increased with RH; the fitted diffusion coefficient initially decreased and then increased above roughly 50% RH. These are model-dependent estimates for this CV study, not measurements assigned automatically to the separate film-density switching study below. Several transport and interfacial mechanisms can coexist.

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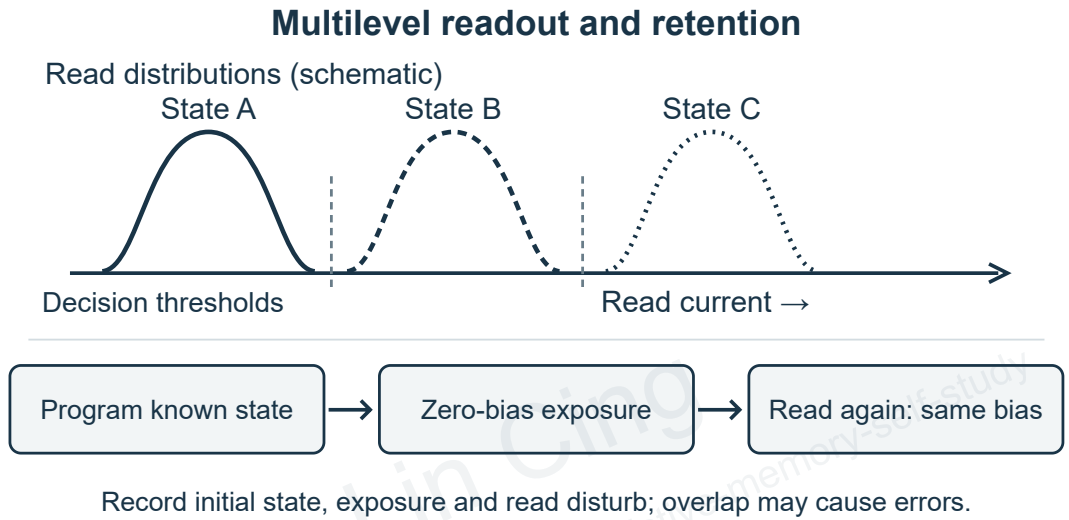
Film density and switching: a separate comparison

In a distinct reported Cu/Ta_2O_5/Pt comparison, films had densities of 7.6 and $7.1\ \mathrm{g/cm^3}$, and forming preceded the I–V measurement at each RH. The example I–V curves include 22% and 85% RH; the $7.6\ \mathrm{g/cm^3}$ sample showed nonvolatile switching under tested RH variation. The $7.1\ \mathrm{g/cm^3}$ sample changed from nonvolatile below 50% RH to volatile above 50%, with recovery when RH fell below 50%.

The authors proposed proton-related conduction as one explanation for the latter behavior and stated that further investigation was needed. Density, deposition and moisture history are coupled; these results do not establish a universal 50% threshold or identify the carrier by I–V shape alone.

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Multilevel readout and retention



Read each programmed state at the same bias before and after the specified zero-bias exposure. Distribution overlap can cause errors even when the nominal current levels differ.

Interpret each evidence chain

In situ plasmon-enhanced spectroscopy of Ag/SiO_2/Au can track spectral changes during switching and support residual particles and changing growth locations. A spectrum is not a direct image: electrode optical response, field enhancement and alternative structures affect interpretation.

For virgin Cu/SiO_2 devices, a parallel-RC fit and area-dependent R and C were reported; a programmed $500 \times 500\,\mu\mathrm{m}^2$ cell spanning about $230\,\Omega$–$55\,\mathrm{k}\Omega$ retained approximately the virgin fitted capacitance. Localized conduction is a plausible interpretation, but equivalent-circuit fits alone cannot identify a unique microscopic path.

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https://github.com/xinhi6716/resistive-memory-self-study

References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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